Brand Purchase Propensity Model - Final Report

1. Overview: Task, Data, and Setup

The task is to predict which of 300 stores will purchase a target brand in a future window, using 15 pre-window behavioral features (`n_orders`, `recency_days`, `target_hist_orders`, etc.). Label prevalence is 16.7% (imbalanced). An evaluation set of 100 stores (no labels) was scored at the end. Primary metric: **PR-AUC**; secondary: **ROC-AUC** and **Precision@Top-10%**, reflecting that campaigns realistically target only a short list of highest-scored stores; all three were confirmed empirically to be threshold-invariant. Correlation analysis showed `target_hist_orders`, `n_orders`, and `orders_per_month` as the strongest individual predictors. Skewed features were `log1p`-transformed for the linear model only (checked empirically, not assumed - one feature, `tenure_days`, was excluded after transforming made its skew worse); tree models used raw features. Given only 300 rows (50 positives), all evaluation used **Repeated Stratified 5-fold CV** (up to $5\times5 = 25$ folds) rather than a single split, to keep the reported metrics stable. Development was assisted by Claude (Anthropic).

2. Limitations

Small sample size drives a CV standard deviation of $\approx$0.10 on PR-AUC, meaning most differences reported below - between models, hyperparameter settings, and engineered features - are not statistically distinguishable from noise; they should be read as suggestive, not conclusive. A learning curve showed validation PR-AUC still rising at the largest training size tested while training PR-AUC saturated near 1.0, indicating the practical ceiling is set by sample size, not model capacity or tuning effort. Extensive model and hyperparameter search on the same 300 rows introduces some optimistic selection bias that cannot be fully corrected after the fact.

3. Methodology: Why These Models Were Tried

The modeling approach followed a deliberate progression rather than a fixed model list. We started with a **regularized Logistic Regression** as an interpretable baseline, appropriate given the small sample size and the need to sanity-check which raw signals actually mattered before adding complexity. We then added **tree ensembles (Random Forest, Gradient Boosting)** because they can capture non-linear feature interactions and require no scaling or distributional assumptions - a good fit for small, mixed-scale tabular data. **Alternative classifiers (KNN, Naive Bayes, SVM)** were tried next purely as a robustness check, to confirm no stronger model family had been overlooked before committing further tuning effort to trees. Once two models (Gradient Boosting and SVM) emerged as comparably strong but structurally different - one axis-aligned/tree-based, one margin-based - **blending** was tested on the reasoning that diverse models make different errors and can reduce variance on a small dataset more reliably than tuning any single model further. Separately, once hyperparameter tuning showed diminishing returns, **feature engineering and data augmentation** were explored as alternative levers to push past the apparent performance ceiling; both were treated as experiments to falsify, not assumed to help.

4. Baseline and Model Comparison

The tuned Logistic Regression (ElasticNet, grid search) reached **PR-AUC 0.452** (ROC-AUC 0.778, P@Top10% 0.393), with 9 of 15 coefficients zeroed by the penalty - concentrating signal on `target_hist_orders`, `recency_days`, and `orders_per_month`. A coarse-grid comparison against Random Forest and Gradient Boosting showed all three within noise of each other (PR-AUC: LogReg 0.448, RF 0.450, GBC 0.436), motivating a deeper search before selecting a final model family.

5. Deeper Hyperparameter Search

A coarse-to-fine `RandomizedSearchCV` (150 candidates, wide continuous ranges for learning rate, subsampling, and leaf/split regularization) was run per model. **Gradient Boosting improved to PR-AUC 0.470** (ROC-AUC 0.782, P@Top10% 0.460), with the winning configuration using heavy stochastic regularization (subsample $\approx$0.61, `min_samples_leaf`=33, learning rate $\approx$0.06, 691 trees) - a "many weak, regularized trees" recipe suited to small n. A properly tuned **sigmoid-kernel SVM** (found via the same coarse-to-fine strategy across rbf/linear/poly/sigmoid kernels) reached **PR-AUC 0.478** (ROC-AUC 0.760, P@Top10% 0.493), outperforming an initial quick RBF-only attempt (0.418) and confirming SVM as a legitimate second candidate alongside GBC.

6. Exploratory Negative Results

Two additional directions were explored and both were negative, which is itself informative: (i) engineering 13 additional ratio/flag features degraded performance when added together, and a controlled one-at-a-time test showed individual effects were smaller than CV noise and did not combine usefully; (ii) SMOTE-based augmentation with a tuned MLP improved over an untuned MLP but never matched the tree/margin models. Both results supported the conclusion that, at $n = 300$, the original 15 features already capture the learnable signal, and further complexity was not productive.

7. Blending

Blending was evaluated with proper nested cross-validation (each base model refit inside every fold to avoid leakage). Individual tuned models scored: LogReg 0.449, RF 0.443, GBC 0.470, SVM 0.478 (PR-AUC). Blending GBC with the weaker RF/LogReg diluted its signal (GBC+RF: 0.462, GBC+LogReg: 0.467, all-four equal weight: 0.475) - all below GBC alone once it had pulled ahead. However, blending the two **strongest and most diverse** models helped: **GBC+SVM (weighted 1:2) reached the best observed PR-AUC of 0.491** (ROC-AUC 0.790, P@Top10% 0.487), and GBC+SVM blends also produced the best ROC-AUC seen in the project (≈0.79 across several weightings).

8. Final Model, Threshold, and Submission

The final model is the **GBC (weight 1) + sigmoid-kernel SVM (weight 2)** blend, with both base models refit on all 300 training rows. Since PR-AUC/ROC-AUC/Precision@Top-10% are threshold-invariant, the required `binary_prediction` column used a **prevalence-matching threshold** (derived from training-only out-of-fold predictions, avoiding any leakage from the evaluation set) rather than an F1-optimal threshold, since the latter implausibly flagged ≈50% of stores against a 17% base rate. The chosen threshold (0.284) produced a 19.0% flagged rate on the evaluation set, closely matching training prevalence, with an out-of-fold sanity-check PR-AUC of 0.497 for the blend. The final `submission.csv` contains `rid`, `p` (blended probability), and `binary_prediction` for all 100 evaluation stores, with checks for duplicate IDs, row-count match, and valid probability range.